

Effects of Strength and Conditioning Training on Athletic Performance in Adolescent Athletes: A Review

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ABSTRACT

The purpose of this study was to evaluate the current body of available literature regarding strength and conditioning in adolescent male and female athletes, describe potential benefits, and generate recommendations for strength and conditioning programs for male and female adolescent athletes. Systematic searches of the PubMed and Google Scholar databases were run using the following keywords Plyometric training, adolescents, strength and conditioning, pubertal physiological changes, strength training, Combined Resistance Training, Counter Movement jump. Studies included in this review evaluated the effectiveness of Strength and Conditioning protocols. Though the critical potential benefits of Strength and Conditioning training, such as improved landing mechanics, coaches and athletic trainers do not routinely implement S&C programs for adolescent athletes. Strength and Conditioning programs have the ability to elevate skills, decrease injury rates, and improve lifelong fitness levels in adolescent athletes. Strength and Conditioning programs for adolescent male and female athletes should incorporate stretching of the hip adductors, targeted hamstring, gluteal and quadriceps strengthening, and a synergistic adaptation model, which tailors training protocols to an athlete's pubertal stage.

Key Words: *Plyometric training, adolescents, strength and conditioning, pubertal physiological changes, strength training, Combined Resistance Training, Counter Movement jump.*

INTRODUCTION

Strength training and conditioning (S&C) is a term that has been adapted to include several modalities of exercise. Strength training via resistance training (RT) serves as the core, and other modalities of exercises are included depending on the needs of the athlete. For example, a Strength and Conditioning

program for strength and power athletes would include weight training but also plyometrics, sprint/agility training, flexibility exercises, and aerobic training (in addition to the rigors of practice and competition). Strength and Conditioning is a simple way to improve athletes' body by losing fat and defining muscles at the

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same time. Strength and Conditioning is about maximum return of your input in a short period of time, with a structural approach to training (Arede, Jet al., 2018).

Athletic performance is based on a number of skills that can be generated through a Strength and conditioning program. This particular program focuses on developing both muscular strength and power using a protocol called post-activation potentiation (PAP), also commonly referred to as complex training.

Complex training combines strength exercises from the load phase and power exercises from the performance phase to improve both muscle force production (strength) and the rate of force production (power). A complex training set contains performing two exercises back-to-back, with rest period in between. The first exercise is a strength or resistance exercise using a heavy weight for four to six repetitions. The second exercise is a power exercise which focuses on explosive movement for 5 to 8 repetitions. A 30- to 45-second rest in between the strength and power exercises should be there and a 90- to 120-second minute rest interval after the completion both exercises.

It is beneficial to perform a number of movement exercises for a proper warm-up before starting a high-intensity training program. There are two protocol to do a S&C workout: Complete complex

repetition of one exercise before going to the next, or combine the exercises into a circular wheel. Circular training allows you to reduce the rest time between complex repetition, which increases the hardening of the workout.

Adolescent Athlete: The athletes of the age group of 12 to 18 years are called adolescent athletes. This age group is very important for future sports career. Children of both genders have the same parameters such as weight, height, injury risks, motor skills, percentage of body fat endurance, strength and haemoglobin levels. After activation of the hypothalamic pituitary-gonadal axis which means puberty occurs, these specific parameters are altered (Faigenbaum, A et al., 2000).

Effect of Puberty on Athletes: Puberty affects every athlete's body irrespective of gender differences. But it affects significantly in the female athletes.

For example: Body fat percentage in normal female body at the time of puberty is 23% to 27% and in the normal male body it is 13% to 15%. As a result of intense exercise and prolonged training female athletes of elite class can decrease this level 12% to 16% (in distance runner) and 8% to 10% (in sprinters). whereas an elite male athlete can reduce their body fat percentage by 4% to 8% in highly trained male gymnasts (Pichardo, A. W., et al., 2019).

Thermoregulatory capacity is

similar in male and female athletes but in pubertal age the heat regulation system is developed. But there is a higher risk of heatstroke in adolescent athletes of both gender if they are late maturing, lower body surface area, or exercise in hot climate.

Female adolescent athletes have better balancing and flexibility, which initiate in childhood and peak at 14 or 15 years of age; in contrast, males improve in flexibility from mid-adolescence until the end of puberty.

Female adolescent athletes develop less strength than male adolescent athlete at the time of puberty. Although the muscle fiber type is similar but the muscle fiber size is less in females. Female develop a small increase in muscle strength after onset of menstruation, whereas the male continues to increase muscle strength throughout the process of puberty (Faigenbaum, A. D. 2000).

The aim of this study was to assess the effects of a strength and conditioning practice on adolescent male and female athletes considering the post-match days and days before the next match on power, acceleration, and change-of-direction speed in players. The novelty of this research consisted in scheduling the training intervention considering post-match days and days before to the next match.

The SJ (Spot Jump) and CMJ (Counter Movement Jump) are simple tests traditionally used to assess leg

power. Research articles also confirmed the likely improvement in SJ and CMJ as a consequence of the combined effect of regular practice and the 8-week strength and conditioning program in comparison with the expected effect of the insulate regular basketball practice. The main adaptations to strength and power training are well documented in those literature, especially those concerning functional (e.g., motor unit coordination) and structural (e.g., muscular hypertrophy).

The increase in leg strength may result by some factors. The pubertal subjects (i.e., Strength and Conditioning group) significantly have lower anabolic hormone concentrations in blood, that makes barrier the muscle hypertrophy, and strength variations are consequence of changes in motor unit synchronization, different neural drive to the musculature, and inhibitory mechanisms. The current results are below the range with adolescent players. However, different methodological designs were explored, specifically training load (i.e., 10-12 repetition maximum), maturational status (i.e., tanner stage 3 or 4) and training background (i.e., no experience). Thus, a synergistic adaptation may have shown, due to increased anabolic hormonal levels in blood which induced changes in muscle architecture, hypertrophic development potentiation for exercise loads selected (10-12 RM) and neural adaptations which dominate in the early

stages in strength conditioning training programs (Nichols, et al., 2001).

The growth and maturity are linked with morphological changes, like increased fascicle length, pennation angle, musculature and tendon's cross-sectional area, tendon stiffness, but also, more muscle unit recruitment, pre-activation, reflex control and decreased co-contraction. Due to these adaptations, that could sustain the improvement in stretch-shortening function, more mature subjects could perform exercises with higher levels of eccentric strain on the muscular unit (Greydanus, D. E., & Patel, D. R. 2002).

Plyometric training exploring horizontal force-vector application improved largely the acceleration performance (5 and 10-meters sprint time), in adolescent basketball players, comparing with vertical training group. The training program was designed that included drills (i.e., repeated jumps, hurdles, ladders, hops) with an emphasis in ankle plantar flexion. The ankle flexors are the most contributors to vertical support forces and increases in stride length at lower speeds (up to 7 m·s⁻¹) and drill specificity may have contributed to the effect on acceleration performance (Ziv, G., & Lidor, R. 2009).

The Pro Agility Test is largely used for assessing the change of direction during team-sports. The S&C group showed higher improvement in change of direction speed comparing to the normal

Adolescent group. The fastest performance in 180° change of direction speed test was mainly described by higher contribution of isometric and eccentric strength capacities, as well as higher braking and propulsive forces, lower contact time, time spent braking and propulsive (Faigenbaum, A., 2000).

The isometric strength seems to be determinant to optimize the triple extension during 180° change of direction speed test, because allows the right alignment of lower limb to reaccelerate. Thus, regular basketball practice plus strength and conditioning intervention is beneficial than regular basketball practice alone. Another finding of those study was to compare the effects of a long-term periodized combined resistance training program with or without weightlifting movements on the motor skill development of adolescent boys. The findings suggest that performance gains were similar between both groups, but the pattern of change varied (Guagliano, J. M., et al., 2013, Hackett, D.A., et al., 2013).

The findings also suggest that movement competency and strength are trainable through periods of the adolescent growth spurt. It was expected that due to the complex and explosive nature of the lifts, the inclusion of weightlifting exercises in a combined resistance training program would induce larger gains in movement competency, strength, power, and sprint performance.

Adaptations to specific training types may also be influenced by maturation status. Some studies in prepubertal children have found that strength, jump, and sprint performance improved more after 12 weeks of weightlifting or plyometric and a high velocity resistance training program than age-matched controls participating in sports only. In addition, 6 weeks of plyometric training elicited greater gains in squat jump and acceleration performance in pre-peak height-velocity children, whereas combined (traditional strength training 1 plyometrics) was most effective for sprint and jump performance in post pubertal children. The initial gain in resistance training skills and absolute and relative strength, as well as the delayed gains in sprint, jump, and throw performance, may have resulted from the long-term periodized nature of the program. Specifically, initial increases in back and front squat strength resulted in improved subsequent sprint performance, suggesting that strength may serve as a basis for power development. The Research articles showed that relatively low-resistance training intensity may be adequate to stimulate moderate strength adaptations in adolescent boys with a relatively low training age (Zebis, M. K., et al., 2016).

However, to promote long-term changes in performance and realize greater magnitudes of training induced neuromuscular adaptation, participants

will need to progressively overload and train with relatively higher resistance training intensities. Furthermore, increases in lower-limb muscular reactivation, stiffness, and power as children mature may contribute to the larger gains in jump and sprint performance seen in the second half of the intervention. The results of those studies demonstrate the importance of establishing movement competency and muscular strength before focusing on power development in adolescent and the specificity of adaptations according to the emphasis of training. Training age may also contribute to the aforementioned lag effect in power and sprint performance due to the type of training performed during different stages of development. For example, previous research has shown that larger strength gains occur during the first year than the second year of resistance training in adolescent male athletes. Furthermore, Till et al. also found that a group of rugby athletes (aged 16–19 years) with a resistance training age of 0 or 2 years saw greater annual gains in strength compared to a group with 1-year training experience. However, the older athletes were likely all post pubertal, so the results may not translate to an adolescent population (Till, K., et al., 2013).

Therefore, they essentially have a higher training age for those tests and therefore take longer to respond to training. More importantly, however, the

more complex activities do not respond as much until training intensity and complexity increase, which primarily occurred in the second half of the periodized program. In periods where training effects were most evident, they are well beyond those expected for growth and maturation. For example, in the 14-week mid- to post-period, CMJ height improved by 0.10% and sprint performance by 3–6% (Waugh, C. M., et al., 2014).

In athletic boys, aged between thirteen and seventeen years completing plyometric or resistance training twice a week for between twelve and sixteen weeks, sprint performance has been reported to increase by 2–6% and CMJ height by 7–9%. Even when considering the entire intervention period and converting to annual rates of improvement, boys in this study improved CMJ height by a rate of 0.26% per year and sprint performance by 0.9% per year. These annual rates of improvement are well beyond those reported for the CMJ (6.9% per year) and sprint performance (2.7–3.1% per year) of boys in a talent development program but who received no resistance training. This study aimed to compare the effects of a combined resistance training with or without weightlifting on the long-term athletic development of adolescent boys (Lesinski, M., et al., 2016, Lloyd, R. S., et al., 2016).

Major findings revealed largely

similar performance gains between both groups. Interestingly, there were clear differences in the pattern of change across variables. In the first half of the year, significant improvements in movement competency and strength were observed, whereas in the second half of the year, jump, throw, and sprint performance significantly improved. Although height and body mass changed at consistent rates throughout the intervention, the different temporal responses shown for the performance variables likely aligned to the periodized design of the training program. Therefore, to effectively make long-term gains in motor skill performance, both coaches and athletes should trust in a long-term training process and adopt patience when pursuing long-term goals (Weakley, J. J., et al., 2019).

Conclusion: Upon reaching puberty, however, female athletes start to fall behind their male peers in both participation rates and training opportunities. Strength and Conditioning programs have the ability to elevate skills, decrease injury rates, and improve lifelong fitness levels in adolescent athletes. A proper Strength and Conditioning protocol should be efficient tailored to a participant's sex and pubertal status. To that end, coaches should be encouraged to pursue higher Strength and Conditioning credentials in order to successfully design and implement safe and effective Strength and Conditioning

programs for both male and female athletes. School districts and administrators may also consider implementing more stringent criteria in the hiring of coaches or athletic trainers

who will be involved with overseeing Strength and Conditioning programs. Changes such as these are crucial to facilitating the success and maximizing the potential of adolescent athlete.

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